Data management I

The labs and practice sets of Day 1. You type your work in the do-file named on each page; this booklet is the printed copy to read, plan and write notes on. Core first, then Extension, then Challenge.

How the checks work. Every task in the do-file ends with a CHECK line. Run the file with Ctrl + D: a silent CHECK means your answer is correct; red text under a CHECK means look at that task again. Never edit a CHECK line. The solutions booklet explains the reasoning behind each answer; read it after you have tried.

Lab 1: First look at the Q1 2026 file

File `C:/NBS_Stata/day1/lab_D1_1.do`
inputs data/lfs_raw.dta
outputs logs/lab_D1_1.log

The situation
Your section head asks three quick questions about the Q1 2026 LFS file before the team starts work on it: how many records, how many variables, how many men and women? Answer them from a do-file that leaves a log, so the answers can be checked by anyone.

Task 1 Settings and the diary — Core

Type these four lines exactly: version 17 clear all cd ”C:/NBS_Stata” log using ”logs/lab_D1_1.log”, replace text

Task 2 Open the survey file — Core

Open lfs_raw.dta from the data folder. Use the short path and the clear option, as in your my_day1.do.

Task 3 How many records? — Core

Count the records in the file. Write the number in the comment: * Records: _____

Task 4 How many men and women? — Core

Make a table of ilo_sex. Write the two counts in the comment: * Men: ____ Women: ____

Task 5 Where do they live? — Extension

Make a table of the region variable. Find its name in the Variables window. Which region has the most records? * Region: ___
___

Task 6 A file that survives a broken run — Challenge

When a run stops halfway, its diary stays open and the next run stops at the log using line. 1. Put a * in front of the last line (log close) and run twice: see the red text on the second run. 2. Add one line near the top so this never blocks a run again. 3. Run twice more: both runs must be clean. 4. Remove the * from the last line. No CHECK line: two clean runs in a row are the check.

A correct lab. the whole file runs from top to bottom with no red text, ends with ”end of do-file”, and logs/lab_D1_1.log exists.

Lab 2: Clean the job variables

File `C:/NBS_Stata/day1/lab_D1_2.do`
inputs data/lfs_teach.dta (your product from Part 9)
outputs logs/lab_D1_2.log

The situation
The employment section asks for clean job variables before it prepares the hours and social-protection tables of the release: hours without codes, labels a reader understands, and a 0/1 indicator for ”employer pays social contributions”.

Task 1 Settings, diary, data — Core

Settings and diary as in every do-file today, then open data/lfs_teach.dta. Log name: logs/lab_D1_2.log

Task 2 Usual hours without codes — Core

wkt_mj_ushrs_rep holds usual weekly hours in the main job. Find the code above 168 (codebook), check its label, and turn it into .a with mvdecode, as you did for actual hours.

Task 3 A label a reader understands — Core

Give wkt_mj_ushrs_rep the variable label: Usual hours, main job (exactly this text).

Task 4 Paid leave: don’t-know codes — Core

mie_pvaclv (paid annual leave) and mie_pscklv (paid sick leave). Read both labels with codebook. If 97 means don’t know, make it .a in both variables with one mvdecode line.

Task 5 A 0/1 indicator for social contributions — Core

Create socsec: 1 if mie_socpro is YES (1), 0 if NO (2), missing otherwise. Hint: generate ... if ... , as for employed. Then label it: Employer pays social contributions (0/1)

Task 6 Part time and full time — Extension

Group actual hours (wkt_mj_acthrs) into hrsgrp: 1 = under 35 hours, 2 = 35 hours or more. Use recode with generate(). Label the two groups ”Under 35” and ”35 or more”.

Task 7 Which region has the highest share? — Challenge

The mean of a 0/1 variable is a share. Use bysort and summarize to find the region with the highest share of employees whose employer pays social contributions (records, unweighted). Write the region here: * Region: _____ Share: ____ No CHECK line: compare your answer with your neighbour’s.

A correct lab. the whole file runs with no red text, ends with ”end of do-file”, and logs/lab_D1_2.log exists. This lab does not save over lfs_teach.dta.

Practice sets

File `C:/NBS_Stata/day1/practice_D1.do`. One block per part of the day; each opens its own data, so the parts can be done in any order. Type your number after `nbscheck` where the file asks for it.

Part 3: Look before you count (lfs_raw.dta, as at the end of Part 3)

P3.1 Core How many records have ilo_lfs19 missing? Use count if. Type the number after nbscheck P3.1.

P3.2 Core List unique_ind ilo_age dem_mrt for records 101 to 105. Type the age of record 103.

P3.3 Core Mean age of men and of women, with bysort. Type the mean age of women, one decimal.

P3.4 Extension How many employed persons work more than 48 actual hours (wkt_mj_acthrs)? Watch the don’t-know code 997: it is not 997 hours.

P3.5 Challenge Import data/district_lookup.csv and count the districts on Mahé (type the é: ”Mahe” finds none). (This replaces

the data in memory: last exercise.)

Part 4: Missing and don’t-know codes (lfs_raw.dta, as at the end of Part 4)

P4.1 Core Read the label of mie_pvaclv with codebook, then turn its don’t-know code into .a with mvdecode.

P4.2 Core misstable summarize for mie_*. Which variable has the most .a? Type how many .a it has.

P4.3 Core Write one assert: ilo_age is never above 85. Your assert is the check: silent means it holds.

P4.4 Extension Find the don’t-know code of usual hours (wkt_mj_ushrs_rep) and turn it into .a.

P4.5 Challenge Write one assert that checks: anyone with a labour force status is at least 15.

Part 6: New variables (lfs_raw.dta, as at the end of Part 6)

P6.1 Core Create female (1 woman, 0 man) and tabulate it.

P6.2 Core Create youth: 1 for ages 15-24, 0 for 25 and over, missing under 15. Tabulate it with the missing.

P6.3 Core kids: the number of children under 15 in each household, on every row.

P6.4 Extension age10: ilo_age in 10-year groups starting at 15 (15-24, 25-34, ... 75+), missing under 15, labelled.

P6.5 Challenge shemp: the share of household members with a labour force status who are employed, on every row (two egen lines, then one generate).